MANGU 2025 KCSE PREDICTION

CONFIDENTIAL EXAM

121/2

MATHEMATICS

PAPER 2

TIME: 2½ HOURS

NAME……………………………………………………………………………………

SCHOOL…………………………………………… SIGN………………………

INDEX NO………………………………………… ADM NO……………………

(Kenya Certificate of Secondary Education)

INSTRUCTIONS TO CANDIDATES

a) *Write your **name, admission number, class and index number** in the space provided.*
b) *This paper contains **TWO** sections: **section I** and **section II***
c) *Answer **ALL** the questions in **Section I** and only **five** questions from **section II.***
d) ***Show all the steps in your calculations, giving your answers at each stage in the spaces provided below each question.***
e) *Marks may be given for correct working even if the answer is wrong.*
f) ***Non-programmable** silent electronic calculators and KNEC mathematical tables may be used except where stated otherwise.*

FOR EXAMINER'S USE ONLY:

Section I

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	TOTAL

Section II

17	18	19	20	21	22	23	24	TOTAL

GRAND TOTAL

SECTION I (50 MARKS)

Answer ALL the questions in this section.

1. Without using logarithms tables or calculator: Evaluate. **(3mks)**

$Log_{10}96 + \frac{3}{4}\ Log_{10}625 - Log_{10}12$

2. Find the value of p if the expression $\boldsymbol{px^2 - 60x + 25}$ is a perfect square, given that p is a constant. **(2mks)**

3. The price of a new car is shs. 800,000. If it depreciates at a constant rate to shs. 550,000 within 4years, find the annual rate of depreciation **to 2d.p.** **(3mks)**

4. Mwangi truncated 7/9 to 3 decimal places. Calculate the percentage error resulting from the truncating. **(3mks)**

5. The position vector of A and B are $\vec{a} = 4\mathbf{i} + 4\mathbf{j} - 6\mathbf{k}$ and $\vec{b} = 10\mathbf{i} + 4\mathbf{j} + 12\mathbf{k}$. D is a point on AB such that AD:DB is 2:1. Find the co-ordinates of D. **(3mks)**

6. A quantity y varies partly as x^2 and partly as x. When y = 6, x = 1 when y = 30, x = 3. Find y when x = - 3. **(3mks)**

7. The coordinates of the end points of diameter are A(2,4) and B(2,6). Find the equation of a circle in the form $ax^2 + by^2 + cx + dy + e = 0$ **(3mks)**

8. A bag M contains 8 balls of which 3 are red and 5 are white. Another bag N contains 7 balls of which 4 are red and 3 are white. A bag is chosen at random and two balls picked at random without replacement. **Using a tree diagram,** Find the probability that the two balls chosen are red in colour. **(4mks)**

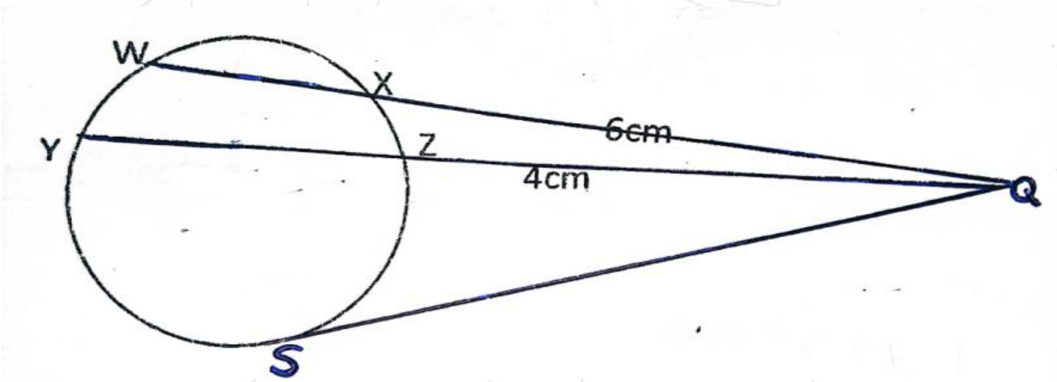
9. Find the equation of tangent to the curve below at the indicated point

$y = x^3 + 3x^2 - 3$ at $x=2$ giving your answer in form of $y = mx + c$

(3mks)

10. Solve the equation $8\sin^2 x = 7 - 2\cos x$ for $0^0 \leq x \leq 360^0$ **(4mks)**

11. Chord WX and YZ intersect externally at Q. The secant WQ= 11cm and QX= 6 cm while ZQ= 4cm.



a) Calculate the length of chord YZ. **(2mks)**

b) Using the answer in (a) above, find the length of the tangent SQ to **2 d.p**. **(2mks)**

12. Find the **possible values** of x given that $\begin{bmatrix} x-1 & x+1 \\ 3x & x \end{bmatrix}$ is a **singular** matrix.

(3mks)

13. Determine the semi interquartile range for the following set of numbers. **(3mks)**

4, 9, 5, 4, 7, 6, 2, 1, 6, 7, 8.

14. **A** carrot patch is in the shape of trapezium. There are 13 carrot plants in the first row, 15 in the second row and 17 in the next and so on. If there are 47 plants in the last row, **how many rows** are there and **how many plants**. **(3mks)**

15. Two brands of coffee cost Kshs 120 and Kshs 144 per kilogram respectively. A wholesaler blends the two and sells the blend at Kshs 168 thereby making a profit of 20%. Find the ratio in which he blends them. **(3mks)**

16. Make x the subject of the formula

(3mks)

$$P^2 = \sqrt{\frac{x + 2W}{4x + 3R}}$$

SECTION II (50 MARKS)

Answer any FIVE questions in this section.

17. (i) Complete the table below, giving the values correct to 2 **decimal places** **(2mks)**

X^0	0^0	15^0	30^0	45^0	60^0	75^0	90^0	105^0	120^0	135^0	150^0	165^0	180^0
$\text{Cos } 2x^0$	1.00		0.50	0.00	-0.50		-1.00	-0.87		0.00		0.87	1.00
$\text{Sin}(x^o+30^0)$	0.50		0.87	0.97	1.00		0.87	0.71	0.50	0.26	0.00		-0.50

(ii) Using the grid provided draw on the same axes the graph of $y=\text{Cos } 2x^0$ and $y = \text{Sin } (x^o+30^0)$ for $0^0 \leq x \leq 180^0$ **(5mks)**

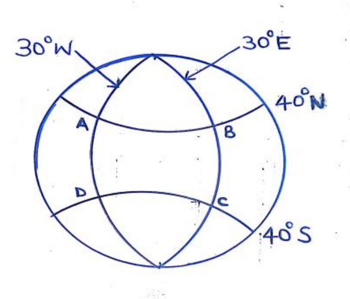
(iii) Find the period of the curve $y=\text{Cos } 2x^0$ **(1mk)**

(iv) Using the graph, estimate the solutions to the following questions to the nearest degree;

a) $\text{Sin}(x^o+30^0) - \text{Cos } 2x^0= 0$ **(1mks)**

b) $\text{Cos } 2x^0=0.50$ **(1mk)**

18. The figure below shows points on the earth's surface.



a) State the positions of A, B, C and D in coordinate form. **(2mks)**

A_________________________

B__________________________

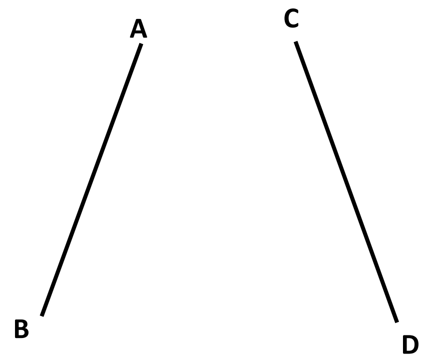
C__________________________

D__________________________

b) An aircraft flies from A to B along latitude 40^0N, B to C along longitude 30^0E, C to D along latitude 40^0S and then flew back to A along longitude 30^0W. Calculate to **2 d.p** the total distance it covered .(**Take radius of the earth =6370 km and $\pi = \frac{22}{7}$**) **(5mks)**

c) If the aircraft leaves A at 8.00am at a speed of 720km/h to B. At what local time is it expected at B? Give your answer **to the nearest minute.** **(3mks)**

19. a) AB and CD are chords of a circle. Construct the circle with centre O and measure its radius **(4mks)**



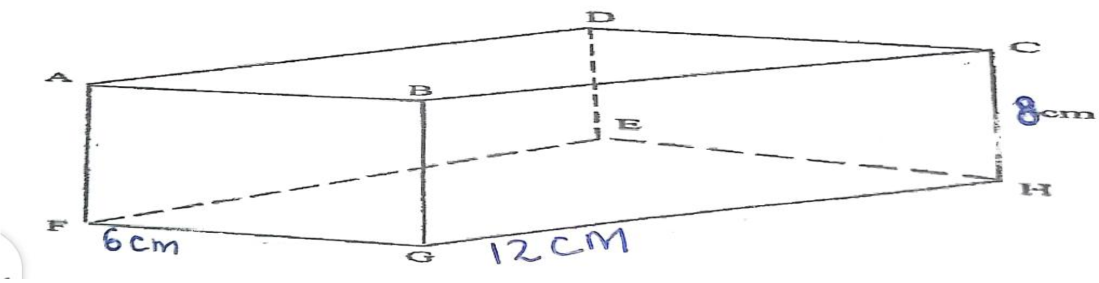
b. Construct the loci of a points x which are equidistant from line AB and CD **(1mk)**

c. Construct the loci of a points Z which are 2cm from the circumference of the circle. **(2mks)**

e. A point P moves such that $CP \geq DP$, It is not more than 2cm from the circumference of the circle and its distance from line CD is not more than its distance from AB. Show the region P by shading it.

(3mks)

20. The diagram below represents a cuboid ABCDEFGH in which FG= 6 cm, GH=12 cm and HC=8 cm



Calculate to **2 decimal places**;

a) The length of FC

(2mks)

b) (i) The size of the angle between the lines FC and FH **(2mks)**

(ii) The size of the angle between the lines AB and FH **(2mks)**

c) The size of the angle between the planes ABHE and the plane FGHE **(2mks)**

d) The space occupied by the Cuboid. **(2mks)**

21. The points A (1,4), B(-2,0) and C (4,-2) of a triangle are mapped onto $A^1(7,4)$, $B^1(x,y)$ and C^1 (10,16) by a transformation N = $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$. Find

(i) Matrix N of the transformation **(4mks)**

(ii) Coordinates of B^1 **(2mks)**

(iii) $A^{II}B^{II}C^{II}$ are the image of $A^1B^1C^1$ under transformation represented by matrix

M = $\begin{pmatrix} 2 & -1 \\ 1 & 0 \end{pmatrix}$ Write down the co-ordinates of $A^{II}B^{II}C^{II}$ **(2mks)**

(iv) A transformation N followed by M can be represented by a single transformation K.

Determine K **(2mks)**

22. A particle moves along a straight line such that its displacement S metres, from a given point is $S = t^3 - 5t^2 + 3t + 4$. Where t is time in seconds find;

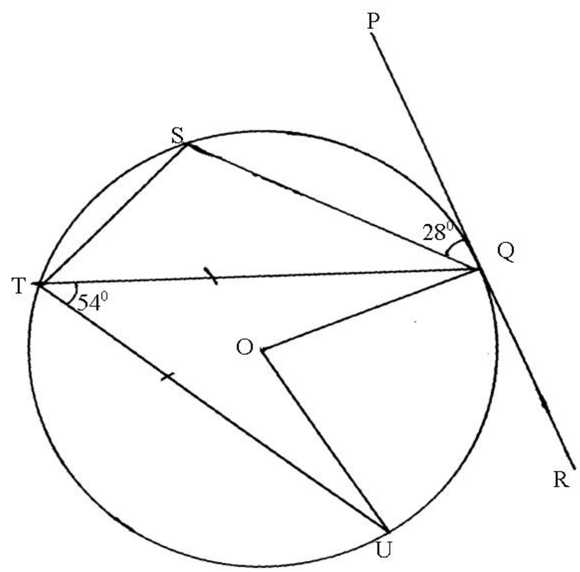
a) The displacement of the particle at $t = 5$ **(2mks)**

b) The velocity of the particle when $t = 5$ **(2mks)**

c) The values of t when the particle is momentarily at rest. **(3mks)**

d) The acceleration of the particle when $t = 2$. **(3mks)**

23. In the figure below ,O is the center of the circle. PQR is a tangent to the circle at Q. Angle PQS=28^0,angle UTQ=54^0 and UT =TQ.



Giving reasons, determine the size of

a) Angle STQ. **(2mks)**

b) Angle TQU. **(2mks)**

c) Angle TQS **(2mks)**

d) Reflex angle UOQ . **(2mks)**

e) Angle TQR. **(2mks)**

24. A company wishes to buy two types of squash machines; Electric and manual. A manual machine requires four attendants whereas an electric one also requires four. An electric machine fills 300 packets per hour; a manual one can fill 200 packets per hour. The numbers of packets to be filled are not more than 3000 per hour and the number of attendants should be at least 40. By letting x be the number of electric Machines and y be the number of manual Machines answer the following questions.

a) Form all the possible inequalities which will represent the above information. **(4mks)**

b) On the grid provided draw the inequalities and shade the unwanted region. **(3mks)**

c) If for every hour it is used, an electric machine brings a profit of shs.200 and a manual one shs.500, determine the number of machines of each type that should be installed in order to maximize profit per hour. Find the maximum profit **(3mks)**